



Artificial Intelligence and Machine Learning

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Department of Computer Applications

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AI Landscape

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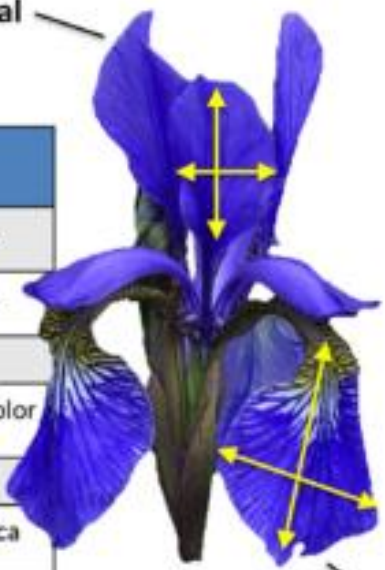
Dataset

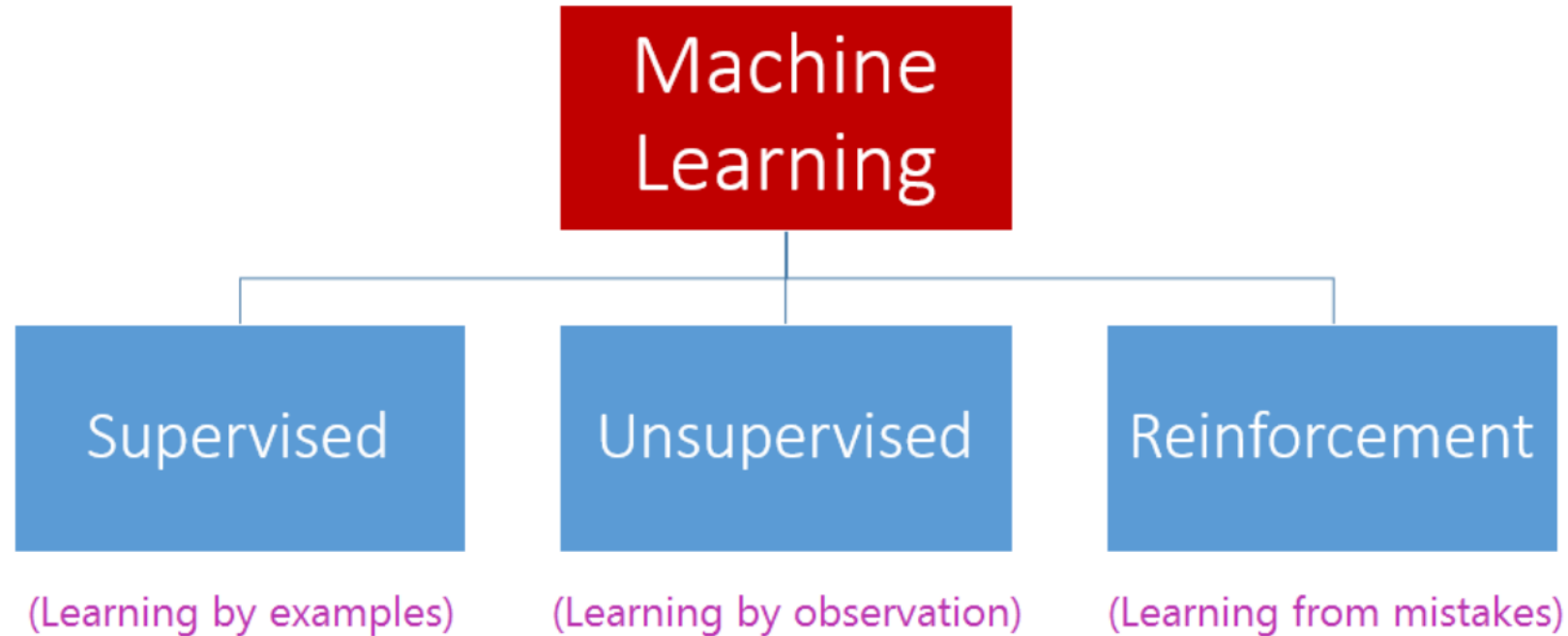
Samples
(instances, observations)

	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

Features
(attributes, measurements, dimensions)

Class labels
(targets)





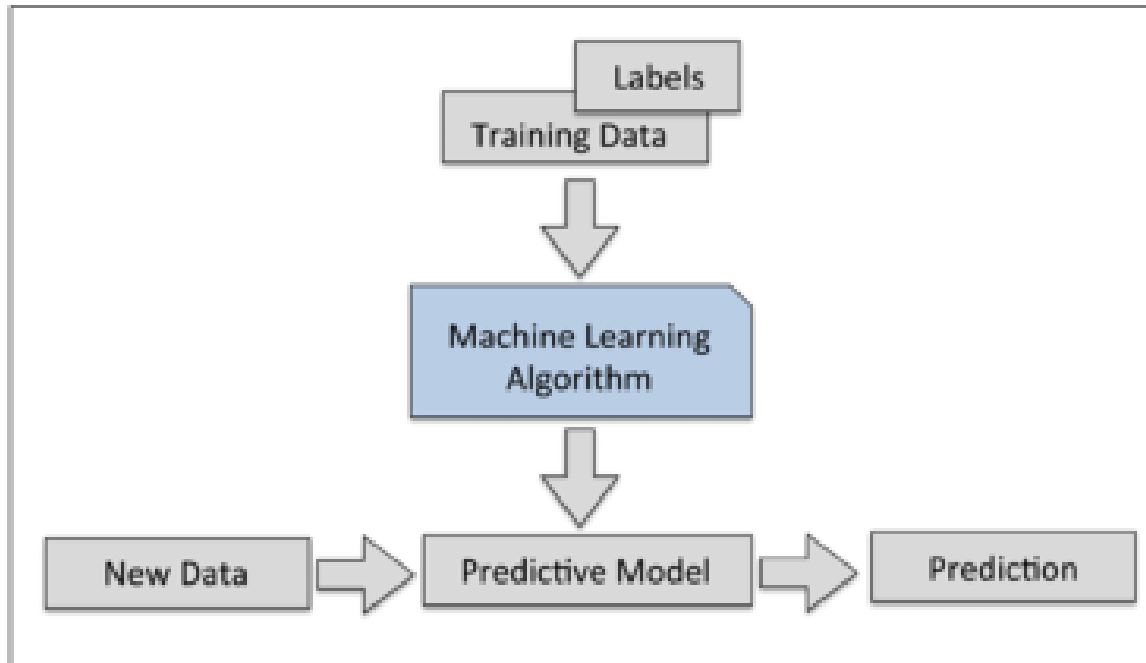
Supervised Learning

Learn a model from labelled training data and Make Predictions about unseen or future data

Supervised – set of samples where the desired output signal is already known

Training data comes with input-output pairs.

Objective: Predict outputs for new inputs based on learned patterns.



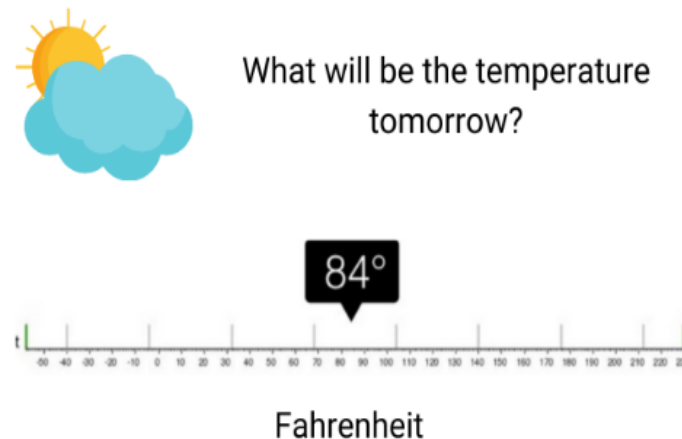
Algorithms: Linear Regression, KNN Classification, Naïve Bayes classification, Support Vector Machines, ANN etc.

Supervised Learning Categories: Classification, Regression

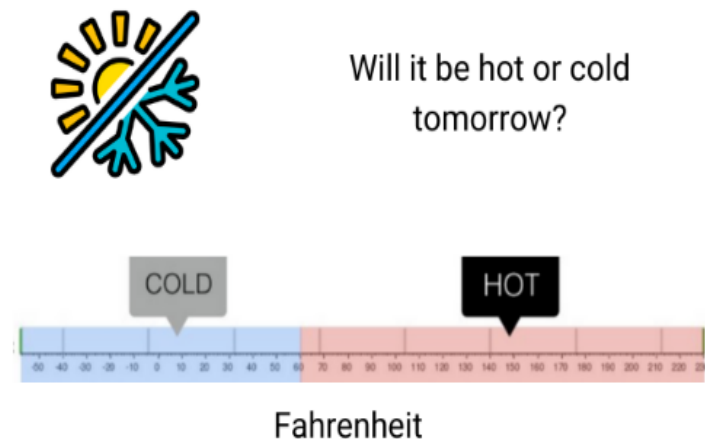
Classification – Predicting **discrete** labels

Regression – Predicting **continuous** values

Regression



Classification



house → "house"

Regression

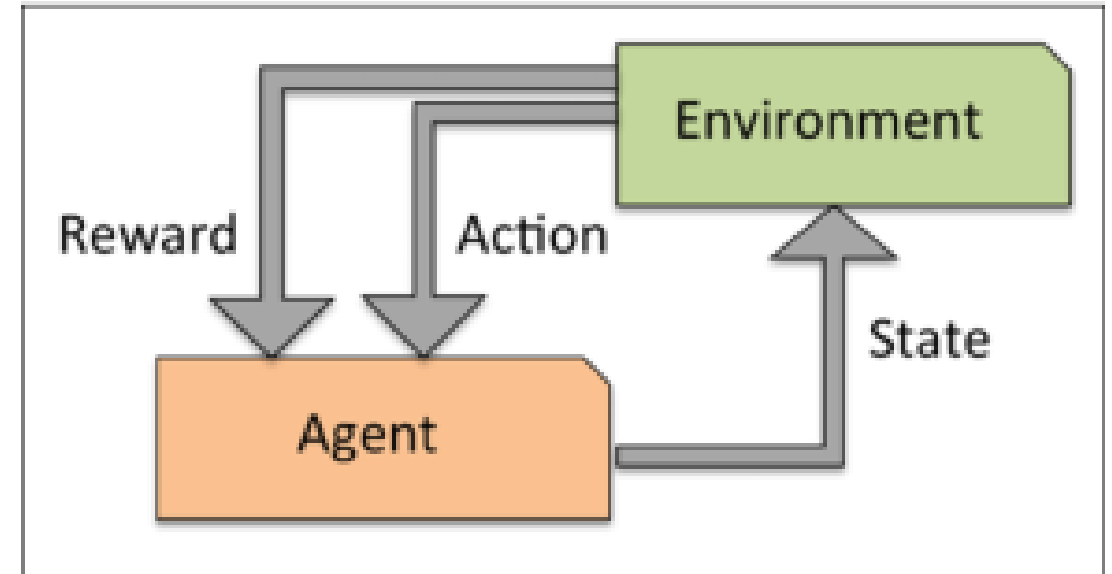


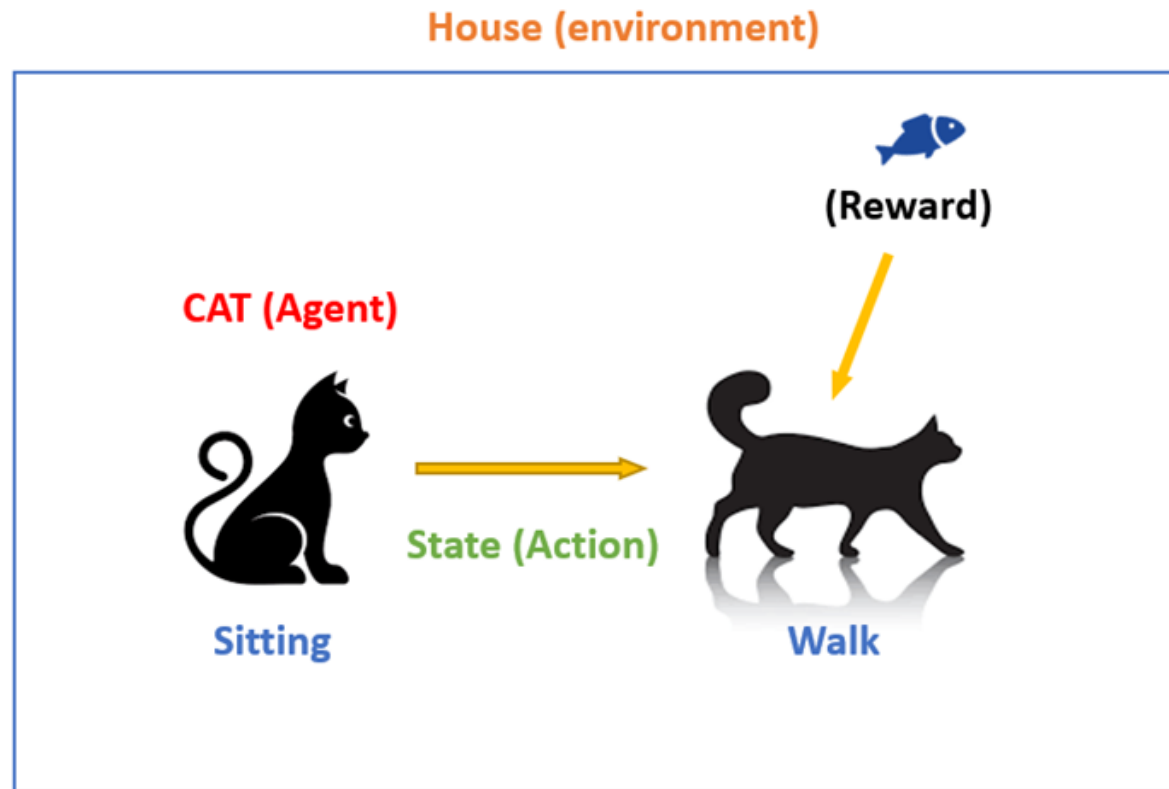
Reinforcement Learning

Goal: Develop a system (agent) which improves its performance based on the interactions with the environment

Maximise the total reward by trial-and-error approach

Algorithms: Q-learning, Deep Q-Networks (DQN), Policy Gradients, Actor-Critic methods.





Unsupervised Learning

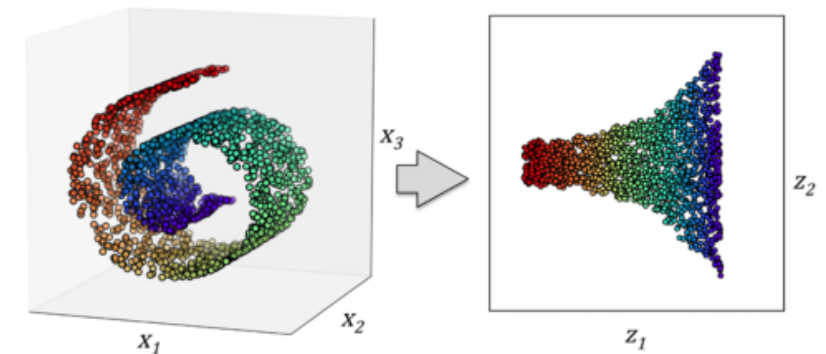
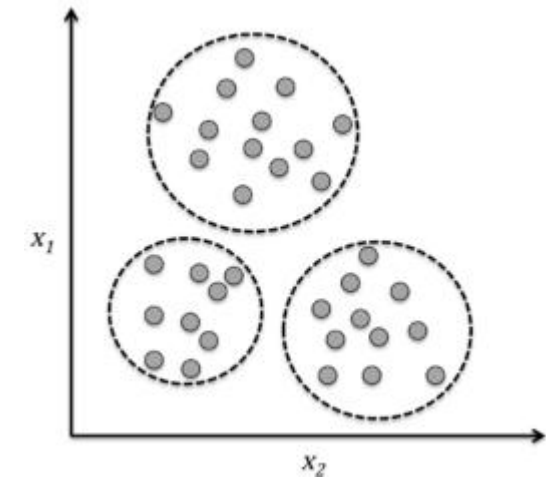
Unsupervised learning involves training a model on data without labeled responses.

Goal: Discover hidden structures or grouping the input data

Clustering - dividing data into meaningful subgroups
Eg: customer segmentation

Dimensionality Reduction: Reducing the number of features thereby reducing the computational complexity

Algorithms: K-means clustering, Hierarchical clustering, Principal Component Analysis (PCA), Association Rule Mining



Spot the Learner - Identify and classify types of Machine Learning

1. A system that predicts house prices based on size, location, and number of rooms.
2. Grouping of customers to target for marketing
3. A robot that learns to walk by getting rewards for every step it takes.
4. Finding patterns in student exam results without any labels.
5. Email spam detection using a labeled dataset.
6. A drone that learns to avoid obstacles through trial and error.



THANK YOU

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